INCH-POUND

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COMMERCIAL ITEM DESCRIPTION

DISPENSER, BULK MILK, MECHANICALLY REFRIGERATED (NAVAL SHIPBOARD)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

This commercial item description (CID) covers self-contained, manually-operated, gravity-fed, mechanically refrigerated cabinets with lift-type or push-type valves. These units are intended to dispense refrigerated milk and fluid milk products from 5-gallon bulk milk containers for Naval shipboard use.

2. CLASSIFICATION

Style 1: Front loading
Style 2: Rear loading

Size 1: One 5-gallon milk container
Size 2: Two 5-gallon milk containers
Size 3: Three 5-gallon milk containers

3. SALIENT CHARACTERISTICS

3.1 Design and construction. The dispenser shall allow the bulk milk containers to be loaded into the refrigerated storage compartment from either the front or rear in accordance with the style specified. The dispenser shall accommodate the size, weight, and load of filled, 5-gallon, dispenser-type milk containers. The minimum interior dimensions and the maximum exterior dimensions of the storage cabinet shall be as specified in table 1. The dispensing mechanism shall allow the milk to flow through the delivery tube in a smooth stream with no dripping after shutoff. No liquid shall remain in the delivery tube on the discharge side of the pinch-off point when the control valve has shut off flow.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: Commander, Naval Sea Systems Command, 2531 Jefferson Davis Highway, Arlington, VA 22242-5160.

AMSC N/A FSC 7310

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TABLE 1. Dimensions for dispenser (inches).

	INTERIOR OF STORAGE COMPARTMENT (MINIMUM)			EXTERIOR OF CABINET (MAXIMUM)		
Dispenser size	1	2	3	1	2	3
Width	11	22	32	16	27	37
Depth	11	11	11	20	20	20
Height	22	22	22	41	41	41

Note: Exterior dimensions include doors, but not door hardware or dispensing mechanisms.

3.2 Performance. The refrigeration system shall be capable of maintaining milk temperatures between 32° and 41 °F in the container and tube while running at ambient temperatures between 65° and 100 °F.

3.3 Cabinet. The dispenser shall consist of a refrigerated storage compartment with an inner liner and outer shell, base assembly, door, legs, hardware, insulation, and panels providing access to the refrigeration system. A thermometer shall be mounted on the front of the cabinet and shall measure the temperature inside the cabinet.

3.3.1 Outer shell. The outer shell shall be stainless steel in accordance with ASTM A167, type 302 or 304 or ASTM A176, type 430, with a thickness of not less than 24-gauge. Removable panels or covers shall be provided as required to provide ready access for servicing of the refrigeration system.

3.3.2 Inner liner. The thickness of the walls and the top of the inner liner shall be not less than 24-gauge stainless steel. The thickness of the liner bottom shall be not less than 18-gauge stainless steel or have an 18-gauge backup plate and shall be fully formed and braced as required to adequately support a full complement of fully-loaded bulk milk containers. It shall withstand, without permanent deflection or distortion, the impact normally associated with the loading operation.

3.3.3 Base assembly. The base assembly shall be constructed to support the fully-loaded storage compartment and to house components of the refrigeration system. The primary framing of the base shall consist of corrosion resistant steel structural shapes or plates with a thickness of not less than 18-gauge.

3.3.4 Door. The door shall consist of a stainless steel inner liner and outer shell of the same thickness and finish as the cabinet inner liner and shell. Door framing shall include suitable plastic or synthetic rubber breaker strips to minimize conduction between the cabinet liner and shell. The door shall be hinged on the right unless left side hinges are specified. Door edge vertical reinforcement shall be 12-gauge corrosion resistant steel. Door hinges and support framing shall be of sufficient strength to prevent misalignment of the door during opening or closing.

3.3.5 Hardware. The door latch, striker plate, hinges, and other miscellaneous hardware shall be stainless steel or chromium-plated metal. A stainless steel door riser shall be provided on the latch side of the size 3 dispenser to ensure door alignment or the door shall house a horizontal structural member to ensure door alignment.

3.3.6 Mounting. The unit shall be provided with four 4-inch high stainless steel legs suitable for secure attachment to a countertop or storage stand.

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3.4 Refrigeration system. The refrigeration system shall consist of a hermetic motor-compressor unit, a static or forced air condenser, a drier, evaporator section with capillary tube or thermal expansion valve and a thermostatic control. Refrigerant piping or tubing shall be securely soldered or welded to the inner liner. Refrigerant systems shall be completely dehydrated, fully charged and tested for leaks prior to shipment.

3.4.1 Refrigerant (ozone depleting substance). The refrigerant shall be nonflammable, be in accordance with ARI 700, and be approved by the Environmental Protection Agency under the Significant New Alternate Program. (It shall have an ozone depletion potential of not greater than 0.05.)

3.5 Drip trays. Drip trays for dispensers shall consist of a shallow drain pan with a removable stainless steel louvered cover. The pan shall be plastic or stainless steel. The drip trays shall be either the full length continuous type designed for placement on the counter or attachment beneath the dispensing station.

3.6 Inclined operation. The unit shall operate satisfactorily, in accordance with the requirements of this CID (such as door remaining closed to prevent loss of contents), when test-operated for 30 seconds inclined at an angle of 15° (30° when specified for submarines) each side of the vertical, in each of two vertical planes at right angles to each other. (Shipboard requirement, see 7.2)

3.7 Environmental suitability. The unit shall be capable of withstanding ship's vibration and motion. Controls, switches, moving parts, and electrical circuits shall operate under shipboard conditions without malfunction, binding, excessive looseness, or damage, when tested in accordance with MIL-STD-167-1, type I equipment. The unit shall be secured to the test machine in the same manner that it will be secured on shipboard. (Shipboard requirement, see 7.2)

3.8 Standards compliance. The unit shall be in accordance with and certified to meet the applicable requirements of ARI 700, NSF 7, NSF 20, and UL 471.

3.9 Electrical requirements. The unit shall operate on 115 VAC, 60 Hz, single-phase power, as defined in DOD-STD-1399, Section 300, and shall have provisions for making direct (hardwired) connections for electric power (2 conductors) and for equipment grounding (1 conductor). All outermost metallic surfaces shall be grounded via the equipment grounding connection. The grounding resistance between any exposed metallic surfaces and the common ground point shall be not greater than 0.1 ohm. Electrical components, other than the hermetically sealed motor, shall be provided in accordance with NEMA 250, Type 13 or equivalent enclosure protection. Metal parts of electrical components and enclosures shall be inherently corrosion resistant or shall be treated and processed for corrosion resistance in accordance with IEEE Standard 45. Mercury-type float switches shall not be used. (Shipboard requirement, see 7.2)

3.10 Physical size limitations. The unit shall pass or shall disassemble so that it can pass through a shipboard access measuring 26 x 66 inches with 8 inch radius corners. When specified for submarines, the cabinet shall pass or shall disassemble so that it can pass through a 30 inch diameter submarine access. The refrigeration system shall not require disassembly to pass through access openings. (Shipboard requirement, see 7.2)

3.11 Accessibility. The unit shall be physically and visually accessible from the front for operation and for normal maintenance with tools, test

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equipment, and replacement parts. No special tools shall be required for access or parts replacement.

3.12 Finish. Stainless steel shall have a type 2B or smoother finish, in accordance with ASTM A480.

3.13 Label plates. The unit shall be provided with a data nameplate and an instruction plate, both attached to the front of the unit. They shall be readily visible during normal operating use and shall not adversely affect the life and utility of the unit.

3.13.1 Data nameplate. The data nameplate shall contain the manufacturer's name, model, serial number, date manufactured, and any other information needed to uniquely identify the unit.

3.13.2 Instruction plate. The instruction plate shall provide instructions for start-up, operation, and shut-down.

4. REGULATORY REQUIREMENTS

The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The product provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market, or the same product that has been delivered to the Government for shipboard use on a previous procurement. The Government reserves the right to require proof of such compliance.

6. PACKAGING

Preservation, packing, and marking shall be as specified in the contract or purchase order.

7. NOTES

7.1 Ordering data.

- Title, number, and date of this CID
- Style and size required
- Right or left hand door swing
- When required, manuals shall be in accordance with ASTM F760

7.2 Shipboard requirement. Whenever a "(Shipboard requirement)" is included in a paragraph under SALIENT CHARACTERISTICS, it is meant that the requirement is something that is not normally offered to the commercial market by the manufacturer.

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7.3 Source of documents.

7.3.1 Military documents. Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
DOD-STD-1399 - Interface Standard for Shipboard Systems Section 300A Electric Power, Alternating

7.3.2 Air-Conditioning & Refrigeration Institute (ARI) Standards. ARI Standards are available from the Air-Conditioning & Refrigeration Institute, 4301 North Fairfax Drive, Suite 425, Arlington, VA 22203.

ARI 700 - Specifications for Fluorocarbons and Other Refrigerants

7.3.3 American Society for Testing and Materials (ASTM) Standards. ASTM Standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohoken, PA 19428-2959.

ASTM A167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
ASTM A176 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet, and Strip
ASTM A480 - General Requirements for Flat-Rolled Stainless and Heat-Resisting Steel Plate, Sheet, and Strip
ASTM F760 - Food Service Equipment Manuals

7.3.4 National Sanitation Foundation International (NSF) Standards. NSF Standards are available from the National Sanitation Foundation International, 3475 Plymouth Road, P.O. Box 130140, Ann Arbor, MI 48113-0140.

NSF 7 - Food Service Refrigerators and Storage Freezers

7.3.5 Underwriters Laboratories (UL) Standards. UL Standards are available from the Underwriters Laboratories Inc., 333 Pfingston Road, Northbrook, IL 60062.

UL 471 - Commercial Refrigerators and Freezers

7.3.6 The Institute of Electrical and Electronic Engineers (IEEE) Standards. IEEE Standards are available from The Institute of Electrical and Electronic Engineers, 445 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331.

IEEE 45 - IEEE Recommended Practice for Electric Installations on Shipboard

7.3.7 The National Electrical Manufacturers Association (NEMA) Standards. NEMA Standards are available from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

NEMA 250 - Enclosures for Electrical Equipment (1000 Volts Maximum)

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7.4 Suggested sources of supply. Manufacturers of products known to meet the requirements of this CID are listed below. However, competition is not limited to these companies.

Nor-Lake Inc.
2nd and Elm Streets
P.O. Box 248
Hudson, WI 54016

Norris Dispenser Company
13430 County Road No. 6
Plymouth, MN 55441-4571

Silver King
Division of Stevens Lee Co.
1600 Xenium Ln, N.
Minneapolis, MN 55441-3706

MILITARY INTERESTS:

Custodians:
Army - GL
Navy - SH

Review activities:
Army - QM1
DSCR - GS
Navy - CG, SA

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA - FSS

Preparing activity:
Navy - SH
(Project 7310-0898)

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